

SARI DOSMA SANTA LUSI MANULLANG

19231105

19.5A.26

SERTIFIKASI KOMPETENSI (DATABASE ADMINISTRATOR)

SPESIFIKASI TABEL

1. TABLE PRODUK

● **Struktur Tabel / Schema**

Nama File	Type	Size	Keterangan
ID	INT		PRIMARY KEY
Nama_produk	Varchar	100	Not null
kode_produk	Varchar	20	Not null
kategori	Varchar	50	Not null
satuan	Varchar	20	Not null
harga_satuan	Decimal	15,2	Not null
stok_minimum	int	20	Not null

● **Isi Data / Records**

Id	Nama_produk	kode_produk	kategori	satuan	harga_satuan	Stok_minimum
1	TV LED 42	TV-001	Elektronik	Unit	2500000	5
2	Laptop Acer	LAP-002	Elektronik	Unit	5000000	5
3	Smartphone Samsung	PHO-003	Elektronik	Unit	2000000	5
4	Kulkas LG	REF-004	Elektronik	Unit	3000000	5
5	Mesin Cuci Panasonic	MES-005	Elektronik	Unit	2200000	5
6	Kertas HVS A4	KRT-001	ATK	Rim	45000	5

2. TABEL TRANSAKSI

- Struktur Tabel / Schema

Nama File	Type	Size	Keterangan
ID	INT		Primary key
Id_produk	Int		Foreign_key
Id_pemasok	int		Foreign_key
Tanggal_transaksi	Date		Not null
Jenis_transaksi	Enum	('masuk','keluar')	
Jumlah	Int	20	
Harga	Decimal	15,2	
no_dokumen	Varchar	50	
Uraian	Varchar	100	
Keterangan	Varchar	255	

● Isi Data / Records

Id	Id_produk	Id_pesanan	tanggal_transaksi	Jenis_transaksi	jumlah	harga	no_dokumen	uraian	keterangan
1	1	2	2026-01-04	masuk	50	-	-	-	-
2	2	2	2026-01-04	masuk	60	-	-	-	-
3	3	1	2026-01-04	masuk	60	-	-	-	-
4	4	1	2026-01-04	masuk	50	-	-	-	-
5	5	1	2026-01-04	masuk	40	-	-	-	-
6	1	-	2026-01-25	keluar	30	-	-	-	-
7	2	-	2026-01-26	keluar	40	-	-	-	-
8	3	-	2026-01-27	keluar	50	-	-	-	-
9	4	-	2026-01-28	keluar	20	-	-	-	-
10	5	-	2026-01-31	keluar	30	-	-	-	-
11	6	-	2026-01-01	-	-	4752 0.00	-	saldo awal	-
12	6	-	2026-01-01	keluar	6	4752 0.00	opname bulanan	Pemakaian Bulan Januari	-
13	6	1	2026-02-15	masuk	10	4500 0.00	SPB Nomor 01/SPB-ATK/I I/2026	Pengadaan	-
14	6	-	2026-02-20	keluar	16	4752 0.00	01/SerahTerimaBarang/II	Pengambilan Bidang CDE	-
15	6	-	2026-02-20	keluar	4	4500 0.00	01/SerahTerimaBarang/II	Pengambilan Bidang CDE	-
16	6	-	2026-12-31	keluar	5	4500 0.00	Opname Bulanan Desember (Akhir Tahun)	Pemakaian Bulan Desember	-

3. TABEL PERSEDIAAN

- Struktur Tabel / Schema

Nama File	Type	Size	Keterangan
ID	INT		Primary_key
Id_produk	INT		Foreign_key
Lokasi_gudang	Varchar	50	Not null
jumlah	Int		Not null
tanggal_masuk	Date		
tanggal_keluar	Date		

- Isi Data / Records

Id	Id_produk	lokasi_gudang	jumlah	tanggal_masuk	tanggal_keluar
1	1	Gudang Jakarta	150	2026-01-04	-
2	2	Gudang Jakarta	200	2026-01-04	-
3	3	Gudang Jakarta	100	2026-01-04	-
4	4	Gudang Jakarta	80	2026-01-04	-
5	5	Gudang Jakarta	70	2026-01-04	-
6	6	Gudang Jakarta	22	2026-01-01	-
7	6	Gudang Jakarta	22	-	2026-01-31
8	6	Gudang Jakarta	16	2026-02-15	-
9	6	Gudang Jakarta	26	-	2026-02-20
10	6	Gudang Jakarta	10	-	2026-02-20
11	6	Gudang Jakarta	6	-	2026-12-31

4. TABEL PEMASOK

- Struktur Tabel / Schema

Nama File	Type	Size	Keterangan
ID	INT		Primary Key
Nama_pemasok	Varchar	30	Not null
Kontak	Varchar	20	Not null

- Isi Data / Records

ID	Nama_pemasok	Kontak
1	PT. Kwangya	0811111111
2	PT. ECII	0898765432

SPESIFIKASI HAK AKSES USER

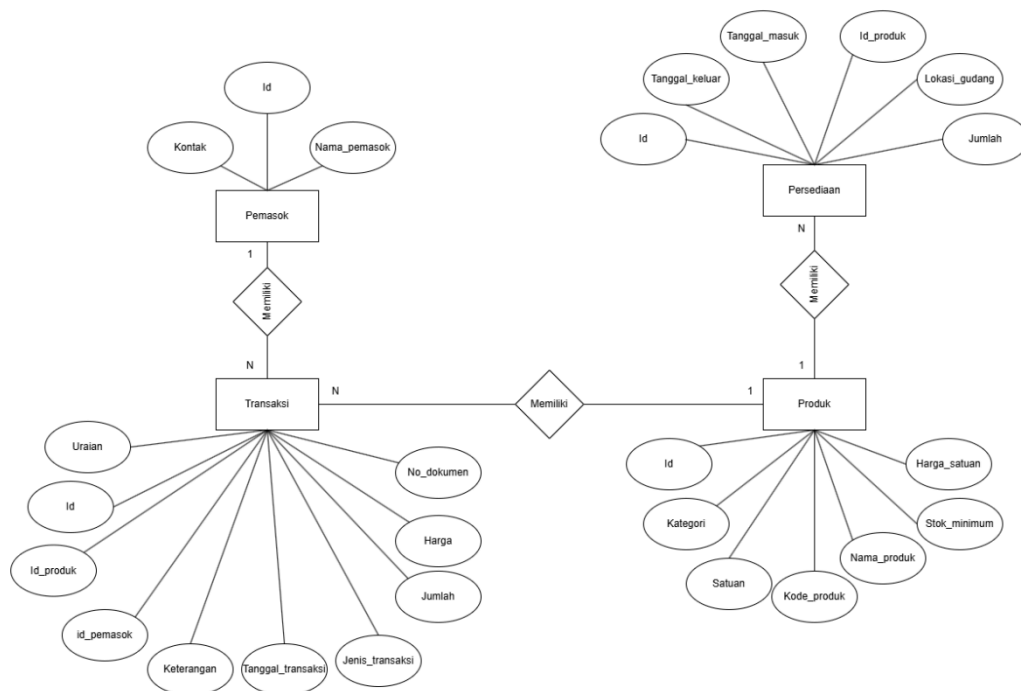
1. ADMIN

USER	PASSWORD	HAK AKSES	TABEL
admin	admin123	ALL PRIVILEGES (SELECT, INSERT, UPDATE, DELETE)	ALL

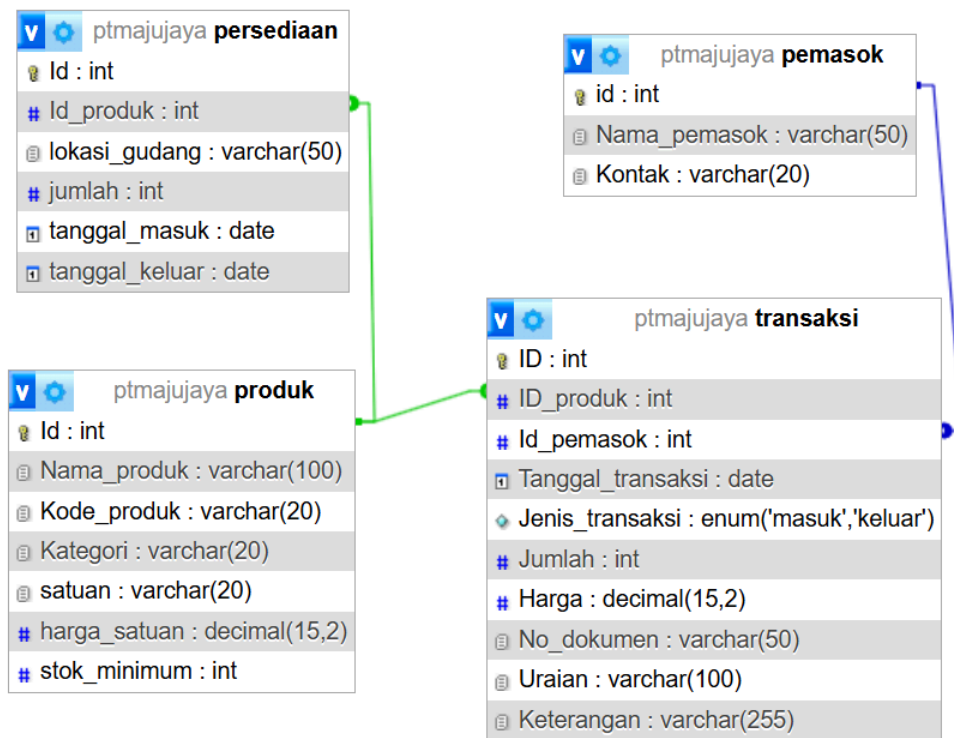
2. KARYAWAN

USER	PASSWORD	HAK AKSES	TABEL
karyawan	1111	SELECT, INSERT, UPDATE	Produk
		SELECT, INSERT, UPDATE	Transaksi
		SELECT, INSERT, UPDATE	Persediaan
		SELECT	Laporanstokgudang
		SELECT	Kartu persediaan barang

ENTITY RELATIONSHIP DIAGRAM (ERD)



LOGICAL RECORD STRUCTURE (LRS)



PEMBUATAN BASIS DATA

1. Login MySQL

```
c:\>mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 48
Server version: 8.0.30 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

2. Membuat database baru

```
mysql> create database PTMajuJaya;
Query OK, 1 row affected (0.01 sec)
```

3. Menggunakan database

```
mysql> use PTMajuJaya;
Database changed
```

4. Membuat struktur tabel produk

```
mysql> create table Produk(
-> Id int auto_increment primary key,
-> Nama_produk Varchar(100) not null,
-> Kode_produk varchar(20) not null,
-> Kategori varchar(20) not null,
-> satuan varchar(20) not null,
-> harga_satuan decimal(15,2) not null,
-> stok_minimum int(20) not null);
Query OK, 0 rows affected, 1 warning (0.05 sec)
```

5. Melihat struktur tabel produk

```
mysql> desc produk;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Id | int | NO | PRI | NULL | auto_increment |
| Nama_produk | varchar(100) | NO | | NULL | |
| Kode_produk | varchar(20) | NO | | NULL | |
| Kategori | varchar(20) | NO | | NULL | |
| satuan | varchar(20) | NO | | NULL | |
| harga_satuan | decimal(15,2) | NO | | NULL | |
| stok_minimum | int | NO | | NULL | |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.01 sec)
```

6. Memasukkan data ke tabel produk

```
mysql> insert into produk (Nama_produk, kode_produk, kategori, satuan, harga_satuan, stok_minimum) values
-> ('TV LED 42', 'TV-001', 'Elektronik', 'Unit', '2500000', '5'),
-> ('Laptop Acer', 'LAP-002', 'Elektronik', 'Unit', '5000000', '5'),
-> ('Smartphone Samsung', 'PHO-003', 'Elektronik', 'Unit', '2000000', '5'),
-> ('Kulkas LG', 'REF-004', 'Elektronik', 'Unit', '3000000', '5'),
-> ('Mesin Cuci Panasonic', 'MES-005', 'Elektronik', 'Unit', '2200000', '5'),
-> ('Kertas HVS A4', 'KRT-001', 'ATK', 'Rim', '45000', '5');
Query OK, 6 rows affected (0.02 sec)
Records: 6 Duplicates: 0 Warnings: 0
```

7. Menampilkan semua data produk

```
mysql> select * from produk;
+-----+-----+-----+-----+-----+-----+
| Id | Nama_produk | Kode_produk | Kategori | satuan | harga_satuan | stok_minimum |
+-----+-----+-----+-----+-----+-----+
| 1 | TV LED 42 | TV-001 | Elektronik | Unit | 2500000.00 | 5 |
| 2 | Laptop Acer | LAP-002 | Elektronik | Unit | 5000000.00 | 5 |
| 3 | Smartphone Samsung | PHO-003 | Elektronik | Unit | 2000000.00 | 5 |
| 4 | Kulkas LG | REF-004 | Elektronik | Unit | 3000000.00 | 5 |
| 5 | Mesin Cuci Panasonic | MES-005 | Elektronik | Unit | 2200000.00 | 5 |
| 6 | Kertas HVS A4 | KRT-001 | ATK | Rim | 45000.00 | 5 |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

8. Membuat struktur tabel transaksi

```
mysql> create table transaksi (
-> ID int auto_increment primary key,
-> ID_produk int not null,
-> ID_pemasok int,
-> Tanggal_transaksi date not null,
-> Jenis_transaksi enum('masuk', 'keluar'),
-> Jumlah int(20),
-> Harga decimal(15,2),
-> No_dokumen varchar(50),
-> Uraian Varchar(100),
-> Keterangan varchar(255),
-> foreign key(Id_produk) references produk(Id),
-> foreign key(Id_pemasok) references pemasok(id));
Query OK, 0 rows affected, 1 warning (0.06 sec)
```

9. Melihat struktur tabel transaksi

```
mysql> desc transaksi;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| ID | int | NO | PRI | NULL | auto_increment |
| ID_produk | int | NO | MUL | NULL | |
| ID_pemasok | int | YES | MUL | NULL | |
| Tanggal_transaksi | date | NO | | NULL | |
| Jenis_transaksi | enum('masuk', 'keluar') | YES | | NULL | |
| Jumlah | int | YES | | NULL | |
| Harga | decimal(15,2) | YES | | NULL | |
| No_dokumen | varchar(50) | YES | | NULL | |
| Uraian | varchar(100) | YES | | NULL | |
| Keterangan | varchar(255) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

10. Memasukkan data ke tabel transaksi

```
mysql> insert into transaksi(id_produk, id_pemasok, tanggal_transaksi, jenis_transaksi, jumlah, harga, no_dokumen, uraian, keterangan) values
-> ('1','2','2026-01-04','masuk','50',null,null,null,null),
-> ('2','2','2026-01-04','masuk','60',null,null,null,null),
-> ('3','1','2026-01-04','masuk','60',null,null,null,null),
-> ('4','1','2026-01-04','masuk','50',null,null,null,null),
-> ('5','1','2026-01-04','masuk','40',null,null,null,null),
-> ('1',null,'2026-01-25','keluar','30',null,null,null,null),
-> ('2',null,'2026-01-26','keluar','40',null,null,null,null),
-> ('3',null,'2026-01-27','keluar','50',null,null,null,null),
-> ('4',null,'2026-01-28','keluar','20',null,null,null,null),
-> ('5',null,'2026-01-31','keluar','30',null,null,null,null),
-> ('6',null,'2026-01-01',null,null,'47520',null,'saldo awal',null),
-> ('6',null,'2026-01-01','keluar','6','47520','Opname bulanan','Pemakaian bulan januari',null),
-> ('6','1','2026-02-15','masuk','10','45000','SPB Nomor 01/SPB-ATK/II/2026','Pengadaan',null),
-> ('6',null,'2026-02-20','keluar','16','47520','01/SerahTerimaBarang/II','Pengambilan Bidang CDE',null),
-> ('6',null,'2026-02-20','keluar','4','45000','01/SerahTerimaBarang/II','Pengambilan Bidang CDE',null),
-> ('6',null,'2026-12-31','keluar','5','45000','Opname Bulanan Desember (Akhir Tahun)','Pemakaian Bulan Desember',null);
Query OK, 16 rows affected (0.01 sec)
Records: 16 Duplicates: 0 Warnings: 0
```

11. Menampilkan semua data transaksi

```
mysql> select * from transaksi;
```

ID	ID_produk	ID_pemasok	Tanggal_transaksi	Jenis_transaksi	Jumlah	Harga	No_dokumen	Uraian	Keterangan
1	1	2	2026-01-04	masuk	50	NULL	NULL	NULL	NULL
2	2	2	2026-01-04	masuk	60	NULL	NULL	NULL	NULL
3	3	1	2026-01-04	masuk	60	NULL	NULL	NULL	NULL
4	4	1	2026-01-04	masuk	50	NULL	NULL	NULL	NULL
5	5	1	2026-01-04	masuk	40	NULL	NULL	NULL	NULL
6	1	NULL	2026-01-25	keluar	30	NULL	NULL	NULL	NULL
7	2	NULL	2026-01-26	keluar	40	NULL	NULL	NULL	NULL
8	3	NULL	2026-01-27	keluar	50	NULL	NULL	NULL	NULL
9	4	NULL	2026-01-28	keluar	20	NULL	NULL	NULL	NULL
10	5	NULL	2026-01-31	keluar	30	NULL	NULL	NULL	NULL
11	6	NULL	2026-01-01	NULL	NULL	47520.00	NULL	saldo awal	NULL
12	6	NULL	2026-01-01	keluar	6	47520.00	Opname bulanan	Pemakaian bulan januari	NULL
13	6	1	2026-02-15	masuk	10	45000.00	SPB Nomor 01/SPB-ATK/II/2026	Pengadaan	NULL
14	6	NULL	2026-02-20	keluar	16	47520.00	01/SerahTerimaBarang/II	Pengambilan Bidang CDE	NULL
15	6	NULL	2026-02-20	keluar	4	45000.00	01/SerahTerimaBarang/II	Pengambilan Bidang CDE	NULL
16	6	NULL	2026-12-31	keluar	5	45000.00	Opname Bulanan Desember (Akhir Tahun)	Pemakaian Bulan Desember	NULL

16 rows in set (0.00 sec)

12. Membuat struktur tabel persediaan

```
mysql> create table persediaan(
-> id int auto_increment primary key,
-> id_produk int not null,
-> lokasi_gudang varchar(50) not null,
-> jumlah int(20) not null,
-> tanggal_masuk date,
-> tanggal_keluar date,
-> foreign key(id_produk) references produk(id));
Query OK, 0 rows affected, 1 warning (0.06 sec)
```

13. Melihat struktur tabel persediaan

```
mysql> desc persediaan;
```

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
id_produk	int	NO	MUL	NULL	
lokasi_gudang	varchar(50)	NO		NULL	
jumlah	int	NO		NULL	
tanggal_masuk	date	YES		NULL	
tanggal_keluar	date	YES		NULL	

6 rows in set (0.00 sec)

14. Memasukkan data ke tabel persediaan

```
mysql> insert into persediaan(id_produk, lokasi_gudang, jumlah, tanggal_masuk, tanggal_keluar) values
-> ('1','Gudang Jakarta','150','2026-01-04',null),
-> ('2','Gudang Jakarta','200','2026-01-04',null),
-> ('3','Gudang Jakarta','100','2026-01-04',null),
-> ('4','Gudang Jakarta','80','2026-01-04',null),
-> ('5','Gudang Jakarta','70','2026-01-04',null),
-> ('6','Gudang Jakarta','22','2026-01-04',null),
-> ('6','Gudang Jakarta','22',null,'2026-01-31'),
-> ('6','Gudang Jakarta','16','2026-02-15',null),
-> ('6','Gudang Jakarta','26',null,'2026-02-20'),
-> ('6','Gudang Jakarta','10',null,'2026-02-20'),
-> ('6','Gudang Jakarta','6',null,'2026-12-31');
Query OK, 11 rows affected (0.01 sec)
Records: 11 Duplicates: 0 Warnings: 0
```

15. Menampilkan semua data produk


```
mysql> select * from persediaan;
```

Id	Id_produk	lokasi_gudang	jumlah	tanggal_masuk	tanggal_keluar
1	1	Gudang Jakarta	150	2026-01-04	NULL
2	2	Gudang Jakarta	200	2026-01-04	NULL
3	3	Gudang Jakarta	100	2026-01-04	NULL
4	4	Gudang Jakarta	80	2026-01-04	NULL
5	5	Gudang Jakarta	70	2026-01-04	NULL
6	6	Gudang Jakarta	22	2026-01-01	NULL
7	6	Gudang Jakarta	22	NULL	2026-01-31
8	6	Gudang Jakarta	16	2026-02-15	NULL
9	6	Gudang Jakarta	26	NULL	2026-02-20
10	6	Gudang Jakarta	18	NULL	2026-02-20
11	6	Gudang Jakarta	6	NULL	2026-12-31

11 rows in set (0.00 sec)

16. Membuat struktur tabel pemasok

```
mysql> create table pemasok(
-> id int auto_increment primary key,
-> Nama_pemasok varchar(50) not null,
-> Kontak varchar(20) not null);
Query OK, 0 rows affected (0.02 sec)
```

17. Melihat struktur tabel produk

```
mysql> desc pemasok;
```

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
Nama_pemasok	varchar(50)	NO		NULL	
Kontak	varchar(20)	NO		NULL	

3 rows in set (0.00 sec)

18. Memasukkan data ke tabel pemasok

```
mysql> insert into pemasok (Nama_pemasok, Kontak) values
-> ('PT.Kwagya','0811111111'),
-> ('PT.ECII','0898765432');
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

19. Menampilkan semua data pemasok

```
mysql> select * from pemasok;
```

Id	Nama_pemasok	kontak
1	PT.Kwagya	0811111111
2	PT.ECII	0898765432

2 rows in set (0.00 sec)

20. Menampilkan laporan stok gudang

a. Membuat Tabel Laporan Stok Gudang

```
mysql> CREATE VIEW laporanstokgudang AS
-> SELECT
-> p.id AS NO,
-> p.nama_produk AS "Nama Produk",
-> p.kode_produk AS "Kode Produk",
->
-> MAX(ps.jumlah) AS "Jumlah Awal Bulan (Unit)",
-> IFNULL(SUM(CASE
-> WHEN t.jenis_transaksi = 'masuk'
-> AND MONTH(t.tanggal_transaksi) = 1
-> THEN t.jumlah ELSE 0 END),0) AS "Pemasukan (unit)",
-> IFNULL(SUM(CASE
-> WHEN t.jenis_transaksi = 'keluar'
-> AND MONTH(t.tanggal_transaksi) = 1
-> THEN t.jumlah ELSE 0 END),0) AS "Pengeluaran (unit)",
-> (MAX(ps.jumlah)
-> +IFNULL(SUM(CASE WHEN t.jenis_transaksi = 'masuk' AND MONTH (t.tanggal_transaksi) = 1 THEN t.jumlah ELSE 0 END),0)
-> -IFNULL(SUM(CASE WHEN t.jenis_transaksi = 'keluar' AND MONTH (t.tanggal_transaksi) = 1 THEN t.jumlah ELSE 0 END),0)
-> ) AS "Jumlah Akhir Bulan (Unit)",
->
-> CONCAT ('Rp', FORMAT(p.harga_satuan,0)) AS "Harga Satuan (Rp)",
-> CONCAT ('Rp', FORMAT(
-> (MAX(ps.jumlah)
-> +IFNULL(SUM(CASE WHEN t.jenis_transaksi = 'masuk' AND MONTH (t.tanggal_transaksi) = 1 THEN t.jumlah ELSE 0 END),0)
-> -IFNULL(SUM(CASE WHEN t.jenis_transaksi = 'keluar' AND MONTH (t.tanggal_transaksi) = 1 THEN t.jumlah ELSE 0 END),0)
-> ) * p.harga_satuan,0)) AS "Nilai Stok Akhir (Rp)"
->
-> FROM produk p
-> JOIN persediaan ps ON p.id = ps.id_produk
-> LEFT JOIN transaksi t ON p.id = t.id_produk
->
-> WHERE p.satuan = 'unit'
-> GROUP BY
-> p.id,
-> p.nama_produk,
-> p.kode_produk,
-> p.harga_satuan
-> ORDER BY p.id;
Query OK, 0 rows affected (0.02 sec)
```

b. Tampilan tabel laporan stok gudang

```
mysql> select * from laporanstokgudang;
```

NO	Nama Produk	Kode Produk	Jumlah Awal Bulan (Unit)	Pemasukan (Unit)	Pengeluaran (Unit)	Jumlah Akhir Bulan (Unit)	Harga Satuan (Rp)	Nilai Stok Akhir (Rp)
1	TV LED 42	TV-001	150	50	30	170	Rp2,500,000	Rp425,000,000
2	Laptop Acer	LAP-002	200	60	40	220	Rp5,000,000	Rp1,100,000,000
3	Smartphone Samsung	PHO-003	100	60	50	110	Rp2,000,000	Rp220,000,000
4	Kulkas LG	REF-004	80	50	20	110	Rp3,000,000	Rp330,000,000
5	Mesin Cuci Panasonic	MES-005	70	40	30	80	Rp2,200,000	Rp176,000,000

5 rows in set (0.02 sec)

21. Menampilkan kartu persediaan barang

a. Membuat tabel kartu persediaan barang

```
mysql> CREATE VIEW kartu_persediaan_barang AS
-> SELECT
-> ROW_NUMBER() OVER (PARTITION BY t.id_produk ORDER BY t.tanggal_transaksi, t.id) AS no,
-> t.id_produk,
-> t.tanggal_transaksi,
-> t.no_dokumen,
-> t.uraian,
-> IFNULL(CASE WHEN t.jenis_transaksi='masuk' THEN t.jumlah ELSE 0 END,0) AS masuk,
-> IFNULL(CASE WHEN t.jenis_transaksi='keluar' THEN t.jumlah ELSE 0 END,0) AS keluar,
-> IFNULL(
-> IFNULL((SELECT IFNULL(p.jumlah,0) FROM persediaan p WHERE p.id_produk=t.id_produk LIMIT 1),0)
-> + SUM(CASE WHEN t.jenis_transaksi='masuk' THEN t.jumlah WHEN t.jenis_transaksi='keluar' THEN -t.jumlah ELSE 0 END)
-> OVER (PARTITION BY t.id_produk ORDER BY t.tanggal_transaksi, t.id)
-> ,0) AS sisa_produk,
-> CONCAT('Rp ', FORMAT(IFNULL(t.harga,0))) AS harga,
-> CONCAT('Rp ', FORMAT(IFNULL(CASE WHEN t.jenis_transaksi='masuk' THEN t.jumlah*t.harga ELSE 0 END,0))) AS bertambah,
-> CONCAT('Rp ', FORMAT(IFNULL(CASE WHEN t.jenis_transaksi='keluar' THEN t.jumlah*t.harga ELSE 0 END,0))) AS berkurang,
-> CONCAT('Rp ', FORMAT(IFNULL(
-> (
-> IFNULL((SELECT IFNULL(p.jumlah,0) FROM persediaan p WHERE p.id_produk=t.id_produk LIMIT 1),0)
-> + SUM(CASE WHEN t.jenis_transaksi='masuk' THEN t.jumlah WHEN t.jenis_transaksi='keluar' THEN -t.jumlah ELSE 0 END)
-> OVER (PARTITION BY t.id_produk ORDER BY t.tanggal_transaksi, t.id)
-> ) * IFNULL(t.harga,0)
-> ,0),0)) AS sisa_rupiah,
-> t.keterangan AS ket
-> FROM transaksi t;
Query OK, 0 rows affected (0.01 sec)
```

b. Tampilan kartu persediaan barang

```
mysql> select * from kartu_persediaan_barang;
```

no	id_produk	tanggal_transaksi	no_dokumen	uraian	masuk	keluar	sisa_produk	harga	bertambah	berkurang	sisa_rupiah	ket
1	1	2026-01-04	NULL	NULL	50	0	200	Rp 2,500,000	Rp 125,000,000	Rp 0	Rp 500,000,000	NULL
2	1	2026-01-25	NULL	NULL	0	30	170	Rp 2,500,000	Rp 0	Rp 75,000,000	Rp 425,000,000	NULL
2	2	2026-01-04	NULL	NULL	60	0	260	Rp 5,000,000	Rp 300,000,000	Rp 0	Rp 1,300,000,000	NULL
2	2	2026-01-26	NULL	NULL	0	40	220	Rp 5,000,000	Rp 0	Rp 200,000,000	Rp 1,100,000,000	NULL
1	3	2026-01-04	NULL	NULL	60	0	160	Rp 2,000,000	Rp 120,000,000	Rp 0	Rp 320,000,000	NULL
2	3	2026-01-27	NULL	NULL	0	50	110	Rp 2,000,000	Rp 0	Rp 100,000,000	Rp 220,000,000	NULL
1	4	2026-01-04	NULL	NULL	50	0	130	Rp 3,000,000	Rp 150,000,000	Rp 0	Rp 390,000,000	NULL
2	4	2026-01-28	NULL	NULL	0	20	110	Rp 3,000,000	Rp 0	Rp 60,000,000	Rp 330,000,000	NULL
1	5	2026-01-04	NULL	NULL	40	0	110	Rp 2,200,000	Rp 88,000,000	Rp 0	Rp 242,000,000	NULL
2	5	2026-01-31	NULL	NULL	0	30	80	Rp 2,200,000	Rp 0	Rp 66,000,000	Rp 176,000,000	NULL
1	6	2026-01-01	NULL	saldo awal	0	0	22	Rp 47,520	Rp 0	Rp 0	Rp 1,045,440	NULL
2	6	2026-01-01	Opname bulanan	Penakaian bulan Januari	0	6	16	Rp 47,520	Rp 0	Rp 285,120	Rp 760,320	NULL
3	6	2026-02-15	SPB Nomor 01/SPB-ATK/II/2026	Pengadaan	10	0	26	Rp 45,000	Rp 450,000	Rp 0	Rp 1,170,000	NULL
4	6	2026-02-20	01/SerahTerimaBarang/II	Pengambilan Bidang CDE	0	16	10	Rp 47,520	Rp 0	Rp 760,320	Rp 475,200	NULL
5	6	2026-02-20	01/SerahTerimaBarang/II	Pengambilan Bidang CDE	0	4	6	Rp 45,000	Rp 0	Rp 180,000	Rp 270,000	NULL
6	6	2026-12-31	Opname Bulanan Desember (Akhir Tahun)	Penakaian Bulan Desember	0	5	1	Rp 45,000	Rp 0	Rp 225,000	Rp 45,000	NULL

16 rows in set (0.00 sec)

c. Menampilkan kartu persediaan barang dengan id_produk = 6

```
mysql> SELECT
-> k.no,
-> k.tanggal_transaksi AS 'Tanggal transaksi',
-> k.no_dokumen AS 'No dokumen',
-> k.uraian AS Uraian,
-> k.masuk AS Masuk,
-> k.keluar AS Keluar,
-> k.sisa_produk AS 'Sisa Produk',
-> k.harga AS Harga,
-> k.bertambah AS Bertambah,
-> k.berkurang AS Berkurang,
-> k.sisa_rupiah AS 'Sisa rupiah',
-> k.ket
-> FROM kartu_persediaan_barang k
-> WHERE k.id_produk = 6
-> ORDER BY k.no;
```

no	Tanggal transaksi	No dokumen	Uraian	Masuk	Keluar	Sisa Produk	Harga	Bertambah	Berkurang	Sisa rupiah	ket
1	2026-01-01	NULL	saldo awal	0	0	22	Rp 47,520	Rp 0	Rp 0	Rp 1,045,440	NULL
2	2026-01-01	Opname bulanan	Penakaian bulan Januari	0	6	16	Rp 47,520	Rp 0	Rp 285,120	Rp 760,320	NULL
3	2026-02-15	SPB Nomor 01/SPB-ATK/II/2026	Pengadaan	10	0	26	Rp 45,000	Rp 450,000	Rp 0	Rp 1,170,000	NULL
4	2026-02-20	01/SerahTerimaBarang/II	Pengambilan Bidang CDE	0	16	10	Rp 47,520	Rp 0	Rp 760,320	Rp 475,200	NULL
5	2026-02-20	01/SerahTerimaBarang/II	Pengambilan Bidang CDE	0	4	6	Rp 45,000	Rp 0	Rp 180,000	Rp 270,000	NULL
6	2026-12-31	Opname Bulanan Desember (Akhir Tahun)	Penakaian Bulan Desember	0	5	1	Rp 45,000	Rp 0	Rp 225,000	Rp 45,000	NULL

6 rows in set (0.00 sec)

22. Menghapus data tertentu berdasarkan 'Nama_produk'

```
mysql> DELETE FROM produk WHERE Nama_produk = 'Buku A4';
Query OK, 1 row affected (0.01 sec)
```

```
mysql> select * from produk;
```

Id	Nama_produk	Kode_produk	Kategori	satuan	harga_satuan	stok_minimum
1	TV LED 42	TV-001	Elektronik	Unit	2500000.00	5
2	Laptop Acer	LAP-002	Elektronik	Unit	5000000.00	5
3	Smartphone Samsung	PHO-003	Elektronik	Unit	2000000.00	5
4	Kulkas LG	REF-004	Elektronik	Unit	3000000.00	5
5	Mesin Cuci Panasonic	MES-005	Elektronik	Unit	2200000.00	5
6	Kertas HVS A4	KRT-001	ATK	Rim	45000.00	5
7	Buku A4	BK-006	ATK	Rim	50000.00	10

7 rows in set (0.00 sec)

```
mysql> select * from produk;
```

ID	Nama_produk	Kode_produk	Kategori	satuan	harga_satuan	stok_minimum
1	TV LED 42	TV-001	Elektronik	Unit	2500000.00	5
2	Laptop Acer	LAP-002	Elektronik	Unit	5000000.00	5
3	Smartphone Samsung	PHO-003	Elektronik	Unit	2000000.00	5
4	Kulkas LG	REF-004	Elektronik	Unit	3000000.00	5
5	Mesin Cuci Panasonic	MES-005	Elektronik	Unit	2200000.00	5
6	Kertas HVS A4	KRT-001	ATK	Rim	45000.00	5

```
6 rows in set (0.00 sec)
```

23. Menampilkan transaksi dengan jenis_transaksi keluar

```
mysql> select * from transaksi where jenis_transaksi = 'keluar';
```

ID	ID_produk	ID_pemasok	Tanggal_transaksi	Jenis_transaksi	Jumlah	Harga	No_dokumen	Uraian	Keterangan
6	1	NULL	2026-01-25	keluar	30	NULL	NULL	NULL	NULL
7	2	NULL	2026-01-26	keluar	40	NULL	NULL	NULL	NULL
8	3	NULL	2026-01-27	keluar	50	NULL	NULL	NULL	NULL
9	4	NULL	2026-01-28	keluar	20	NULL	NULL	NULL	NULL
10	5	NULL	2026-01-31	keluar	30	NULL	NULL	NULL	NULL
12	6	NULL	2026-01-01	keluar	6	47520.00	Opname bulanan	Pemakaian bulan januari	NULL
14	6	NULL	2026-02-20	keluar	16	47520.00	01/SerahTerimaBarang/II	Pengambilan Bidang CDE	NULL
15	6	NULL	2026-02-20	keluar	4	45000.00	01/SerahTerimaBarang/II	Pengambilan Bidang CDE	NULL
16	6	NULL	2026-12-31	keluar	5	45000.00	Opname Bulanan Desember (Akhir Tahun)	Pemakaian Bulan Desember	NULL

```
9 rows in set (0.00 sec)
```

24. Mengupdate Berdasarkan Id

```
mysql> UPDATE produk
-> SET Nama_produk = 'Pensil 2B',
-> Kode_produk = 'PSL-007',
-> satuan = 'Lusin'
-> WHERE id = 8;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> select * from produk;
```

ID	Nama_produk	Kode_produk	Kategori	satuan	harga_satuan	stok_minimum
1	TV LED 42	TV-001	Elektronik	Unit	2500000.00	5
2	Laptop Acer	LAP-002	Elektronik	Unit	5000000.00	5
3	Smartphone Samsung	PHO-003	Elektronik	Unit	2000000.00	5
4	Kulkas LG	REF-004	Elektronik	Unit	3000000.00	5
5	Mesin Cuci Panasonic	MES-005	Elektronik	Unit	2200000.00	5
6	Kertas HVS A4	KRT-001	ATK	Rim	45000.00	5
8	Buku A4	BK-006	ATK	Rim	50000.00	10

```
7 rows in set (0.00 sec)
```

```
mysql> select * from produk;
```

ID	Nama_produk	Kode_produk	Kategori	satuan	harga_satuan	stok_minimum
1	TV LED 42	TV-001	Elektronik	Unit	2500000.00	5
2	Laptop Acer	LAP-002	Elektronik	Unit	5000000.00	5
3	Smartphone Samsung	PHO-003	Elektronik	Unit	2000000.00	5
4	Kulkas LG	REF-004	Elektronik	Unit	3000000.00	5
5	Mesin Cuci Panasonic	MES-005	Elektronik	Unit	2200000.00	5
6	Kertas HVS A4	KRT-001	ATK	Rim	45000.00	5
8	Pensil 2B	PSL-007	ATK	Lusin	50000.00	10

```
7 rows in set (0.00 sec)
```

25. Mengganti nama kolom/struktur

```
mysql> ALTER TABLE pemasok
-> RENAME COLUMN kontak TO kontak_info;
Query OK, 0 rows affected (0.03 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> desc pemasok;
```

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
Nama_pemasok	varchar(50)	NO		NULL	
kontak_info	varchar(20)	NO		NULL	

```
3 rows in set (0.00 sec)
```

26. Membuat user yang terdiri dari admin dan karyawan

```
mysql> CREATE USER 'admin'@'localhost' IDENTIFIED BY 'admin123';
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> CREATE USER 'karyawan'@'localhost' IDENTIFIED BY '1111';
Query OK, 0 rows affected (0.01 sec)
```

27. Menampilkan user dari mysql.user

```
mysql> select user, host from mysql.user;
+-----+-----+
| user      | host      |
+-----+-----+
| admin     | localhost |
| karyawan  | localhost |
| mysql.infoschema | localhost |
| mysql.session | localhost |
| mysql.sys  | localhost |
| root      | localhost |
+-----+-----+
6 rows in set (0.00 sec)
```

28. Memberikan hak akses ke admin

```
mysql> GRANT SELECT, UPDATE, INSERT, DELETE ON ptmajujaya.* to 'admin'@'localhost';
Query OK, 0 rows affected (0.01 sec)
```

29. Menampilkan hak akses admin

```
mysql> SHOW GRANTS FOR 'admin'@'localhost';
+-----+-----+
| Grants for admin@localhost |
+-----+-----+
| GRANT USAGE ON *.* TO 'admin'@'localhost' |
| GRANT SELECT, INSERT, UPDATE, DELETE ON 'ptmajujaya'.* TO 'admin'@'localhost' |
+-----+-----+
2 rows in set (0.00 sec)
```

30. Memberikan hak akses ke karyawan

```
mysql> GRANT SELECT, UPDATE, INSERT ON ptmajujaya.produk to 'karyawan'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> GRANT SELECT, UPDATE, INSERT ON ptmajujaya.persediaan to 'karyawan'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> GRANT SELECT, UPDATE, INSERT ON ptmajujaya.transaksi to 'karyawan'@'localhost';
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> GRANT SELECT ON ptmajujaya.laporanstokgudang TO 'karyawan'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> GRANT SELECT ON ptmajujaya.kartu_persediaan_barang TO 'karyawan'@'localhost';
Query OK, 0 rows affected (0.00 sec)
```

31. Menampilkan hak akses karyawan

```
mysql> SHOW GRANTS FOR 'karyawan'@'localhost';
+-----+-----+
| Grants for karyawan@localhost |
+-----+-----+
| GRANT USAGE ON *.* TO 'karyawan'@'localhost' |
| GRANT SELECT ON 'ptmajujaya'.'kartu_persediaan_barang' TO 'karyawan'@'localhost' |
| GRANT SELECT ON 'ptmajujaya'.'laporanstokgudang' TO 'karyawan'@'localhost' |
| GRANT SELECT, INSERT, UPDATE ON 'ptmajujaya'.'persediaan' TO 'karyawan'@'localhost' |
| GRANT SELECT, INSERT, UPDATE ON 'ptmajujaya'.'produk' TO 'karyawan'@'localhost' |
| GRANT SELECT, INSERT, UPDATE ON 'ptmajujaya'.'transaksi' TO 'karyawan'@'localhost' |
+-----+-----+
6 rows in set (0.00 sec)
```

32. Menerapkan semua hak akses setiap user

```
mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.00 sec)
```

33. Pengujian hak akses karyawan

- Login sebagai user karyawan

```
c:\>mysql -u karyawan -p
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 49
Server version: 8.0.30 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

- Menampilkan database yang dalam karyawan dan menggunakan database ptmajujaya.

```
mysql> show databases;
+-----+
| Database |
+-----+
| information_schema |
| performance_schema |
| ptmajujaya |
+-----+
3 rows in set (0.00 sec)

mysql> use ptmajujaya;
Database changed
```

- Melakukan insert dalam produk

```
mysql> insert into produk (nama_produk, kode_produk, kategori, satuan, harga_satuan, stok_minimum) values
-> ('Iphone 20', 'IP-007', 'Elektronik', 'Unit', '10000000', '6');
Query OK, 1 row affected (0.01 sec)
```

- Melakukan update pada produk dengan id = 6

```
mysql> UPDATE produk
-> SET kode_produk = 'KRT-006'
-> WHERE id = 6;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

- Tampilan setelah di insert dan diupdate

```
mysql> select * from produk ;
+----+-----+-----+-----+-----+-----+-----+
| Id | Nama_produk | Kode_produk | Kategori | satuan | harga_satuan | stok_minimum |
+----+-----+-----+-----+-----+-----+-----+
| 1 | TV LED 42 | TV-001 | Elektronik | Unit | 2500000.00 | 5 |
| 2 | Laptop Acer | LAP-002 | Elektronik | Unit | 5000000.00 | 5 |
| 3 | Smartphone Samsung | PHO-003 | Elektronik | Unit | 2000000.00 | 5 |
| 4 | Kulkas LG | REF-004 | Elektronik | Unit | 3000000.00 | 5 |
| 5 | Mesin Cuci Panasonic | MES-005 | Elektronik | Unit | 2200000.00 | 5 |
| 6 | Hertas HVS A4 | KRT-006 | ATK | Rim | 45000.00 | 5 |
| 9 | Iphone 20 | IP-007 | Elektronik | Unit | 10000000.00 | 6 |
+----+-----+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)
```

- Melakukan delete produk tetapi gagal karena karyawan tidak ada hak akses delete

```
mysql> DELETE FROM produk WHERE produk_id = 9;
ERROR 1142 (42000): DELETE command denied to user 'karyawan'@'localhost' for table 'produk'
```

34. Penguji hak akses admin

- Login sebagai admin dan menggunakan database ptmajujaya

```
c:\>mysql -u admin -p
Enter password: *****
Welcome to the MySQL monitor. Commands end with ; or \g.
Your MySQL connection id is 50
Server version: 8.0.30 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> use ptmajujaya;
Database changed
```

- Melakukan delete pada produk

```
mysql> DELETE FROM produk WHERE id = 9;
Query OK, 1 row affected (0.01 sec)

mysql> select * from produk;
+----+-----+-----+-----+-----+-----+-----+
| Id | Nama_produk | Kode_produk | Kategori | satuan | harga_satuan | stok_minimum |
+----+-----+-----+-----+-----+-----+-----+
| 1 | TV LED 42 | TV-001 | Elektronik | Unit | 2500000.00 | 5 |
| 2 | Laptop Acer | LAP-002 | Elektronik | Unit | 5000000.00 | 5 |
| 3 | Smartphone Samsung | PHO-003 | Elektronik | Unit | 2000000.00 | 5 |
| 4 | Kulkas LG | REF-004 | Elektronik | Unit | 3000000.00 | 5 |
| 5 | Mesin Cuci Panasonic | MES-005 | Elektronik | Unit | 2200000.00 | 5 |
| 6 | Hertas HVS A4 | KRT-006 | ATK | Rim | 45000.00 | 5 |
+----+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

35. Membuat backup database

- Membuat database baru

```
Microsoft Windows [Version 10.0.26200.7623]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Sari DSL Manullang>mysqldump -u root -p ptmajujaya > backup.sql
Enter password:
```

```
mysql> create database pt_maju_jaya2;
Query OK, 1 row affected (0.01 sec)
```

- Melakukan export data menggunakan mysqldump

```
C:\Users\Sari DSL Manullang>mysql -u root -p pt_maju_jaya2 < backup.sql
Enter password:

C:\Users\Sari DSL Manullang>dir backup.sql
Volume in drive C is OS
Volume Serial Number is 58CF-564B

Directory of C:\Users\Sari DSL Manullang

09/02/2026  15:17          10.051 backup.sql
               1 File(s)              10.051 bytes
               0 Dir(s)  123.333.128.192 bytes free

C:\Users\Sari DSL Manullang>mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 13
Server version: 8.0.30 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> use pt_maju_jaya2;
Database changed
mysql> show tables;
+-----+
| Tables_in_pt_maju_jaya2 |
+-----+
| laporanstokgudang       |
| pemasok                 |
| persediaan              |
| produk                  |
| transaksi                |
+-----+
5 rows in set (0.00 sec)
```